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Assignment #3

Mathematics - III (MTH 231)

Date: September 12, 2024

Branch - Electrical Engineering

Section - I and II

1. What conditions must a system of linear equations satisfy for the Jacobi method or Gauss Siedel method to converge to a solution?
2. Use Jacobi method to solve the following system of equations with $x^{(0)} = (1, 1, 1)^T$ as an initial approximation correct to 2 significant figures.

$$x - 10y + 3z = 39$$

$$10x - 2y - 5z = 26$$

$$4x - 5y + 10z = 47$$

What is the minimum number of iterations required to get 5 significant digit accuracy?

3. Do three iterations of Jacobi method to solve

$$-2x + 3y + 10z = 22$$

$$10x + 2y + z = 9$$

$$x + 10y - z = -22$$

with $x^{(0)} = (1, -1, 1)^T$ as the starting vector. What is the minimum number of iterations required, so that the solution is correct to 4 decimal places?

4. Solve the given system by Gauss-Seidel method correct upto 4 significant figures.

$$3x + 9y - 2z = 11$$

$$4x + 2y + 13z = 24$$

$$4x - 2y + z = -8$$

5. Compute the solution of the following system of linear equations by Gauss-Seidel method, correct upto 3 significant figures.

$$6.7x + 1.1y + 2.2z = 20.5$$

$$3.1x + 9.4y - 1.5z = 22.9$$

$$2.1x - 1.5y + 8.4z = 28.8$$

6. Use Gauss elimination method to solve the system

$$1.660x + 0.684y + 0.820z + 0.380u = -4.925$$

$$0.784x + 1.690y + 1.396z + 0.492u = 6.105$$

$$0.754x + 1.602y + 1.608z + 0.456u = 7.325$$

$$0.442x + 0.570y + 0.338z + 1.398u = -4.175$$

7. Apply Gauss elimination method to show that the given system of linear equations has no solution.

$$\begin{aligned}2 \sin(x) - \cos(y) + 3 \tan(z) &= 3 \\4 \sin(x) + 2 \cos(y) - 2 \tan(z) &= 10 \\6 \sin(x) - 3 \cos(y) + \tan(z) &= 9\end{aligned}$$

8. Let $\mathcal{P}_2$ denote the set of all polynomials with real coefficients and degree ≤ 2 . We need to find all polynomials in $\mathcal{P}_2$ that satisfies

$$f(1) = 2 \qquad \text{and} \qquad f(-1) = 6$$

Formulate this problem as a system of linear equations and use Gauss elimination method to find all the polynomials.

Hint : What does a general polynomial $f(x)$ in $\mathcal{P}_2$ look like?

9. Perform five iterations each of Jacobi as well as Gauss Seidel method to solve the given system with starting initial guess as $(x, y, z) = (1, 1, 1)^T$.

$$\begin{aligned}2x + 3y + 7z &= 16 \\3x + y + z &= 6 \\x + 5y + 3z &= 10\end{aligned}$$

What is the minimum number of iterations required, so that the solutions are correct to 8 significant figures?

10. Does the following system of linear equations converge for Jacobi iteration scheme for every initial solution x_0 ? If yes, justify your answer.

$$\begin{aligned}-2x - 2y - 4z &= 1 \\-2x - y + 2z &= 2 \\4x + 2y + 5z &= 3\end{aligned}$$

11. Check whether the following system of linear equations is diagonally dominant or not.

$$\begin{aligned}2.6x_1 - 3.5x_2 - 0.5x_3 &= 17.3 \\1.1x_1 - 1.4x_2 - 45.7x_3 &= 7.4 \\2.4x_1 - x_2 - 7.7x_3 &= 0.4\end{aligned}$$

12. For the matrices

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 1 \\ 6 \\ 4 \end{bmatrix}$$

find the inverse of A using Gauss Jordan method and hence solve the system of linear equations given by $AX = B$ where $X = (x, y, z)^T$.

13. A rod is rotating in a plane about one of its ends. If the following table gives the angle θ (in radians) through which the rod has turned at different values of time t (in seconds), find its angular velocity at $t = 0.7$ seconds.

t	0	0.2	0.4	0.6	0.8	1
θ	0	0.12	0.48	0.10	2.0	3.20

14. From the following table, find x correct to two decimal places for which y is maximum. Also find the maximum value of y .

x	1.2	1.3	1.4	1.5	1.6
y	0.9320	0.9636	0.9855	0.9975	0.9996

15. In a machine, a slider moves along a fixed straight rod. Its distance x along the rod is given for various values of time t .

- (a) Find the velocity of the slider at $t = 0.3$ seconds.
(b) Find the acceleration of the slider at $t = 0.3$ seconds.

t	0	0.1	0.2	0.3	0.4	0.5	0.6
x	3.013	3.162	3.207	3.364	3.395	3.381	3.324

16. The elevation above a datum line of seven points of a road are given below

Position, x	0	300	600	900	1200	1500	1800
Elevation, y	135	149	157	183	201	205	193

Find the gradient $\frac{dy}{dx}$ of the road at the middle point.

17. Find the force of mortality $u_x = -\frac{1}{l_x} \frac{dl_x}{dx}$ at $x = 50$ years using the following data

Years, x	50	51	52	53
Mortality, l_x	73499	72724	71753	70599

18. Use Stirling's Formula to find the first derivative of the function $y = 2e^x - x - 1$ at $x = 0.6$ taking the values of x at 0.4, 0.5, 0.6, 0.7, 0.8. Find the error of approximation.

19. Estimate the annual rate of cloth sales of 1935 from the following data

Year	1920	1925	1930	1940
Sales (in lakhs of metres)	250	285	328	444

20. Estimate $\int_5^6 \sin \left(\cos \left(e^{5x^2-7} \right) \right)$ taking 13 data points using

- (a) Trapezoidal Rule

- (b) Simpson's $\frac{1}{3}$ Rule
- (c) Simpson's $\frac{3}{8}$ Rule
- (d) Weddle's Rule

21. Calculate the value of $\int_0^{\frac{5}{3}} \frac{x}{1+x} dx$ with three significant figures using ten intervals by Trapezoidal Rule.

22. The velocity of a fluid flowing through a pipe of uniform cross-section is given by $v = x^3 \log_{10}(x)$. Find the distance covered by the fluid in 2 seconds taking $h = 0.25$.

23. A tank is discharging water through an orifice at a depth of x metres below the surface of the water. The cross-sectional area of the tank is $A \text{ m}^2$. Given below are the values of x for corresponding values of A .

A	1.257	1.390	1.520	1.650	1.809	1.962	2.123	2.295	2.462	2.650	2.827
x	1.50	1.65	1.80	1.95	2.10	2.25	2.40	2.55	2.70	2.85	3.00

Using the formula $T = \int_{1.5}^{3.0} \frac{A}{\sqrt{x}} dx$, calculate the time T , in seconds for the level of the water to drop from 3 metres to 1.5 metres above the orifice.

24. Compute the integral $\int_0^{\frac{\pi}{2}} \sqrt{1 - 0.162 \sin^2 \phi} d\phi$ using Weddle's rule.